MEDVISION / DASHBOARD GUIDE

Read the dashboard. Understand the risk.

A practical manual for exploring readmission analytics, understanding model performance, reviewing fairness context, and trying a new encounter.

What MEDVISION estimates

The app estimates the likelihood that an encounter is followed by hospital readmission within 30 days. A score is a model estimate, not a confirmed outcome or clinical decision.

A quick map of the app

HOME	HISTORICAL	NEW DATA
Overview	**Known labels**	**Unknown outcome**
Metrics and interactive charts.	Held-out records with estimates and observed outcomes.	Compact form and a separate model.

Model in this manual: balanced Logistic Regression. Historical evaluation: 20,488 held-out encounters from 14,304 patients. The compact New Data form uses a separate 13-input model. Dashboard values can change if local model artifacts were updated.

For demonstration and learning. Not for diagnosis, treatment, or clinical triage.

01 / DASHBOARD MAP

Start here: what you are looking at

MEDVISION has two prediction contexts. Keep them separate when explaining the app:

Area	What is scored	Is actual outcome known?
Historical Dataset	Previously recorded held-out encounters	Yes. Observed readmission label is available for evaluation.
New Patient Risk Prediction	Form describing a new encounter	No. The future outcome is unknown; only a model estimate is shown.

The target: 30-day readmission

From the dataset's **readmitted** field: **<30 = positive (1)**; **NO** and **>30 = negative (0)**. The model estimates the chance of the positive class from encounter features.

Dataset note

The source is the Diabetes 130-US Hospitals dataset with ICD-9-era diagnosis codes. Some external briefs mention ICD-10; this dataset does not contain ICD-10.

Recommended reading order

- **Home:** understand the selected model and its metrics.
- **Historical:** compare estimates with known outcomes; select a record for explanation and cohort context.
- **New Data:** enter encounter-time information; the outcome is not yet known.

A score is not certainty

A displayed risk of 41.8% is the model's estimated probability under learned patterns. It does not mean readmission is certain and is not a diagnosis.

02 / MODEL PERFORMANCE

Home: understand the performance cards

The current full historical model is balanced Logistic Regression. Test metrics below use a patient-level held-out set and classification cutoff 0.52.

ACCURACY	PRECISION	RECALL
68.88%	**17.96%**	**47.44%**
All encounters classified correctly.	Of positive flags, share actually positive.	Of actual positives, share detected.

F1 SCORE	AVERAGE PRECISION	ROC-AUC
26.05%	**20.95%**	**64.31%**
Harmonic balance of precision and recall.	Precision-recall summary across cutoffs.	Ranking ability across cutoffs.

Four possible outcomes

Term	Meaning
True positive (TP)	A positive readmission case the model flags positive.
False negative (FN)	A positive case the model misses; recall measures how often positives are found.
False positive (FP)	A negative case flagged positive; this lowers precision and can create extra follow-up.
True negative (TN)	A negative case correctly left unflagged.

Why accuracy can mislead

About 11.55% of test encounters are positive. Predicting every record negative would be about 88.45% accurate while detecting none (0% recall). Compare recall, precision, F1, and AUC with accuracy.

Average precision summarizes precision-recall across cutoffs. **ROC-AUC** measures ranking quality; neither means the percentage of predictions correct.

03 / READING THE VISUALS

Cutoffs, risk bands, and charts

These settings do different jobs:

Setting	Value	Purpose
Full-model cutoff	0.52 (52%)	At/above this probability the full model predicts positive.
Compact-model cutoff	0.49 (49%)	Used by the separate New Data model.
Display bands	0-30%, 30-70%, 70-100%	Lower, Moderate, Higher groups used in visuals.

Band is not cutoff

A score of 52% is at the full-model cutoff and can be labeled positive. It remains in the Moderate display band (30%-70%). The band is a visual grouping; the cutoff converts probability to a yes/no predicted label.

Home charts

Chart action	How to read it
Pie selector	Choose Risk group, Diabetes medication, or Insulin treatment to view encounter distribution.
Click a sector	Updates companion bars for that selected subset.
Companion bars	Compare Actual readmission, Average model risk, and Higher-risk share.
Validation snapshot	Uses validation data at cutoff 0.50; metric cards use test data at cutoff 0.52, so values can differ.

Check group counts when comparing slices. Small groups can have unstable rates. Higher-risk share means share in the Higher display band, not observed readmission rate.

04 / KNOWN OUTCOMES

Historical Dataset: review an encounter

This page shows held-out encounters with both a model estimate and observed outcome. It is for reviewing examples and errors, not scoring a new patient.

Item	How to read it
Record	Encounter identifier or row label.
Risk level	Lower, Moderate, or Higher display band.
Age / gender / admission type	Encounter attributes for context.
Actual outcome	Known target label: under-30-day readmission or not under this definition.
Estimated risk	Model probability for the positive class.

Review workflow

- Scan or sort by estimated risk to inspect high-scored encounters.
- Select a row and compare estimated risk with actual outcome. A high score can be a false positive; a low score can be a false negative.
- Read the explanation and fairness context as review aids, not proof of cause or fairness.
- Use overall metrics to judge performance; one record cannot establish model quality.

Historical predictions have labels

The source dataset includes the observed outcome, so the app can compare estimates with labels. A new encounter has no known outcome.

The current test view has **20,488 encounters from 14,304 patients**. Patient-level splitting keeps one patient's encounters on one side of the train/test boundary, reducing leakage. The holdout is smaller than the complete source dataset.

Explanations and fairness cards

Selected-record explanation

The explanation compares inputs with typical training-data values and identifies model signals associated with higher or lower scores. These are learned associations, not causal explanations. They do not prove that a feature caused an individual's readmission.

Fairness analysis

Age, gender, and race cards summarize model behavior for demographic cohorts in the historical holdout. Depending on the card, they show cohort size, observed readmission rate, recall, false-alarm rate, or average estimated risk, often compared with the overall group.

Measure	Meaning
Group size (n)	Number of encounters behind the group summary.
Observed readmission rate	Share with a positive recorded target in the cohort.
Recall	Share of that cohort's actual positives flagged.
False-alarm rate	Share of actual negatives flagged positive.
Average predicted risk	Mean model score for the cohort.
Difference vs overall	How far a group measure is above or below overall.

Read subgroup comparisons carefully

A positive difference only means the measure is higher than its reference; it is not automatically better or worse. Small groups make results unstable. These descriptive cards are not fairness certification and do not explain why disparity exists.

New Data also shows peer-group context from the compact model's historical evaluation cohort. It cannot show the new encounter's actual outcome or establish individual fairness.

06 / UNKNOWN OUTCOME

New Data: enter an encounter

The compact form is an additional dashboard capability. It uses a separate model trained on 13 fields to stay practical; it does not ask for every full-model feature.

Input group	Fields	Entry guidance
Demographics	Age band, Gender, Race	Choose a category; use Unknown when appropriate.
Encounter	Admission type/source, Hospital days	Use information available at prediction time.
Prior use	Prior inpatient, emergency, outpatient visits	Enter requested prior encounter counts.
Clinical summary	Number of diagnoses, medications	Use encounter counts; avoid guessing if unavailable.
Diabetes care	A1C result, Insulin	Choose listed category or Unknown if offered.

Steps

- Open New Data / New Patient Risk Prediction.
- Fill the compact fields; numeric controls show allowed ranges.
- Use Unknown instead of inventing a value when available.
- Select Predict Risk. Read probability, display band, cutoff-based label, and explanation together.

Privacy and timing

The form does not need patient names, phone numbers, addresses, or direct identifiers. Enter only requested features. Avoid information that would only become known after prediction time.

Compact-model snapshot: cutoff 0.49; about 63.65% accuracy, 16.67% precision, 53.65% recall, 25.43% F1, 0.6329 ROC-AUC, and 0.2045 average precision. These are separate from Home's full-model cards and may change after retraining.

07 / REFERENCE

Demo workflow and quick glossary

A short demonstration

- On Home, explain the under-30-day target and class imbalance.
- Explain precision and recall together. The 0.52 full-model cutoff is an operating point, not a medical boundary.
- In Historical Dataset, select one encounter and compare estimate with known outcome.
- Show explanation and fairness cards as descriptive context; associations are not causes and small cohorts are uncertain.
- In New Data, enter a sample and state clearly its future outcome is unknown.

Quick glossary

Term	Plain-language definition
Encounter	One hospital visit/admission record; a patient can have multiple encounters.
Target / label	The recorded value learned by the model: here, under-30-day readmission.
Probability / risk score	Model output between 0 and 1, shown as a percentage.
Threshold / cutoff	Probability boundary for assigning positive or negative predicted label.
Holdout test set	Patient-separated records reserved to estimate performance on data not used to fit the model.
Prevalence	Share of a group with the positive target.
Class imbalance	Positive and negative target classes occur at different rates.
Fairness disparity	Difference in an outcome or performance measure across demographic groups; requires context.

Limits

This educational prototype uses historical 1999-2008 data. Practices, populations, and coding can change. Results do not establish current hospital performance or patient benefit. Do not use the dashboard to make care decisions.

If your local artifacts differ, follow the values in the running dashboard and metrics file. See the project README for setup; train the model first if required artifacts are missing.